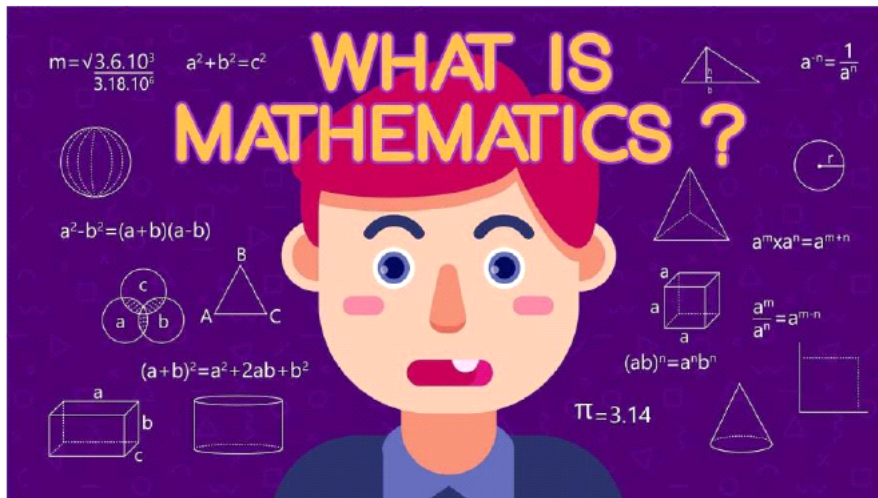


Lec-01-Week-01

Thursday, 13 August 2026 12:50 pm

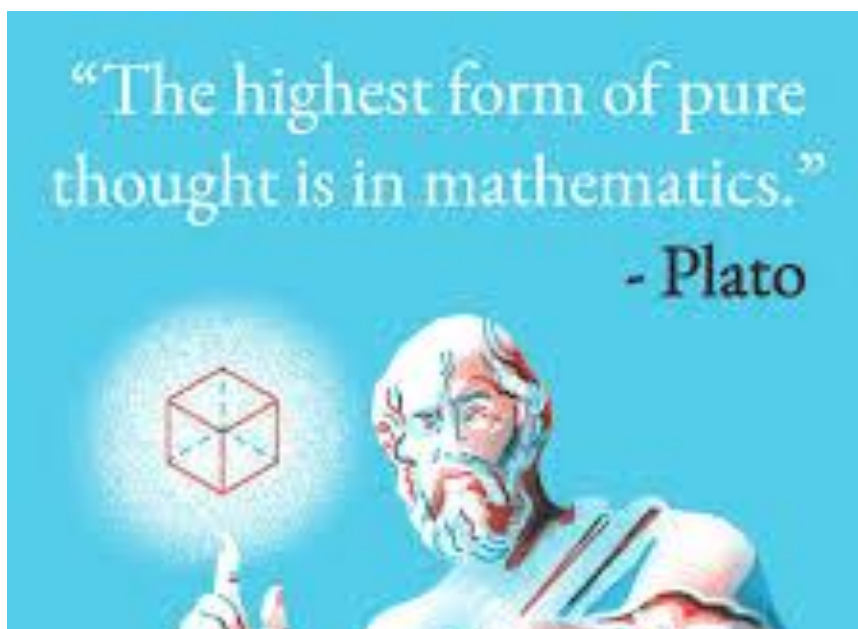
- **Mathematics**

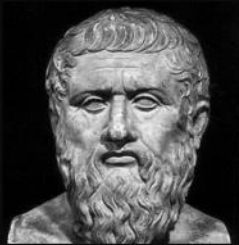


The word *mathematics* comes from [Ancient Greek](#) *máthēma* (*μάθημα*), meaning "that which is learnt", "what one gets to know", hence also "study" and "science".

Its [adjective](#) is *mathēmatikós* (*μαθηματικός*), meaning "related to learning" or "studious", which likewise further came to mean "mathematical".

Similarly, one of the two main schools of thought in [Pythagoreanism](#) was known as the *mathēmatikoi* (*μαθηματικοί*)—which at the time meant "learners" rather than "mathematicians" in the modern sense.

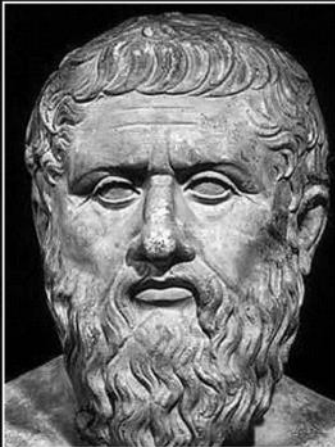




Mathematics is the language in which the gods talk to people.

~ Plato

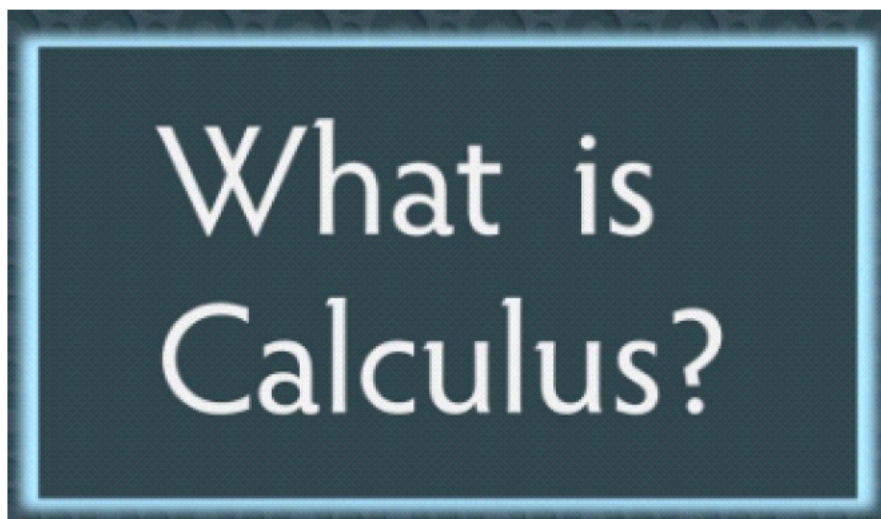
AZ QUOTES



Let no-one ignorant of geometry enter. Said to have been inscribed above the door of Plato's Academy.

— Plato —

AZ QUOTES



Calculus, originally called **infinitesimal calculus** or "**the calculus of [infinitesimals](#)**", is the **[mathematical](#)** study of **continuous change**.

Infinitesimal calculus was developed independently in the late 17th century by [Isaac Newton](#) and [Gottfried Wilhelm Leibniz](#).

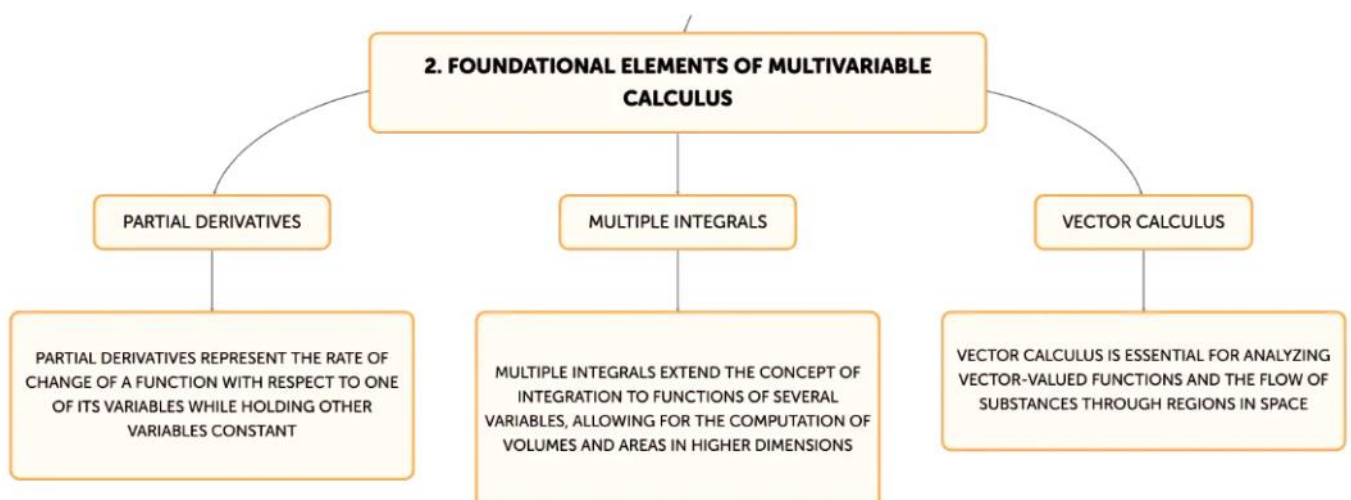
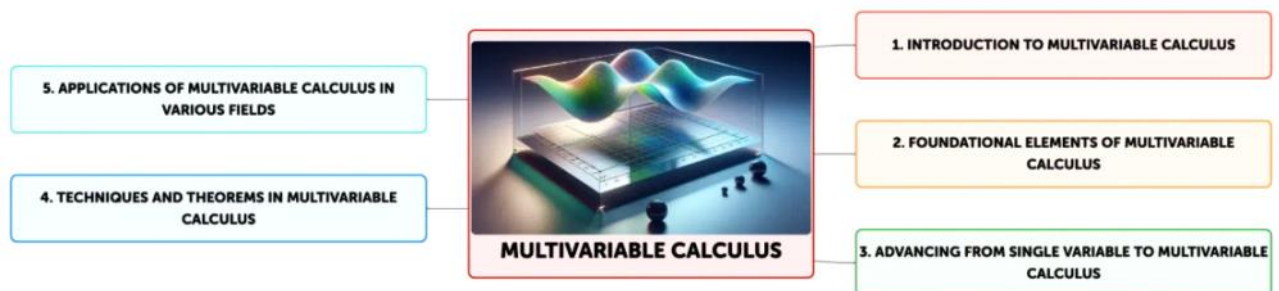


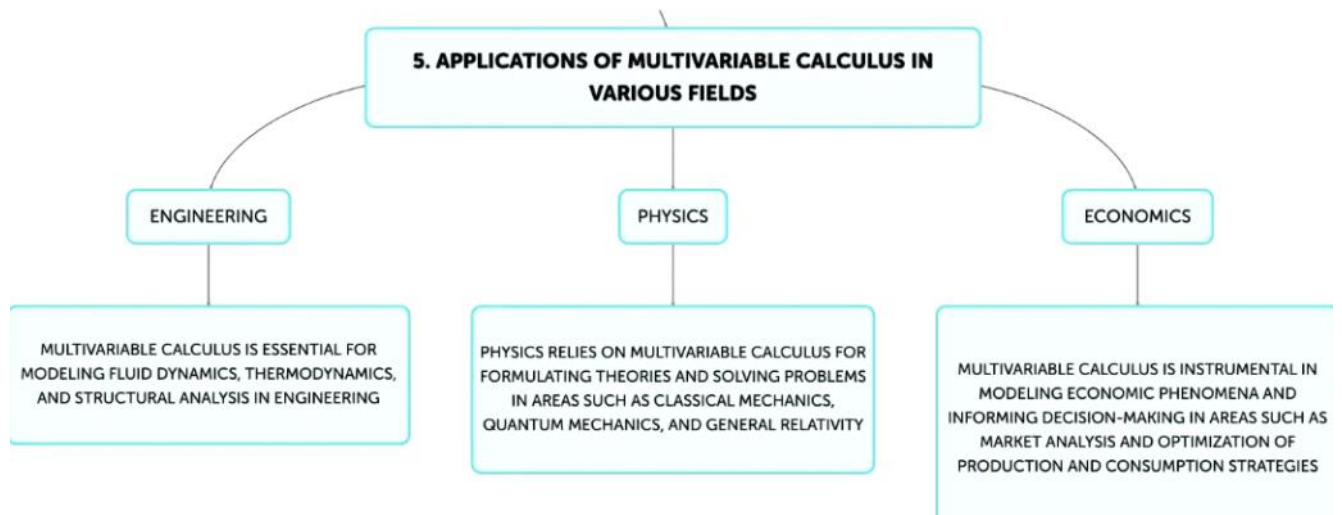
Gottfried Wilhelm Leibniz was the first to state clearly the rules of calculus.



Isaac Newton developed the use of calculus in his laws of motion and universal gravitation.

<https://cards.algoreducation.com/en/content/VO0tc885/multivariable-calculus-overview>

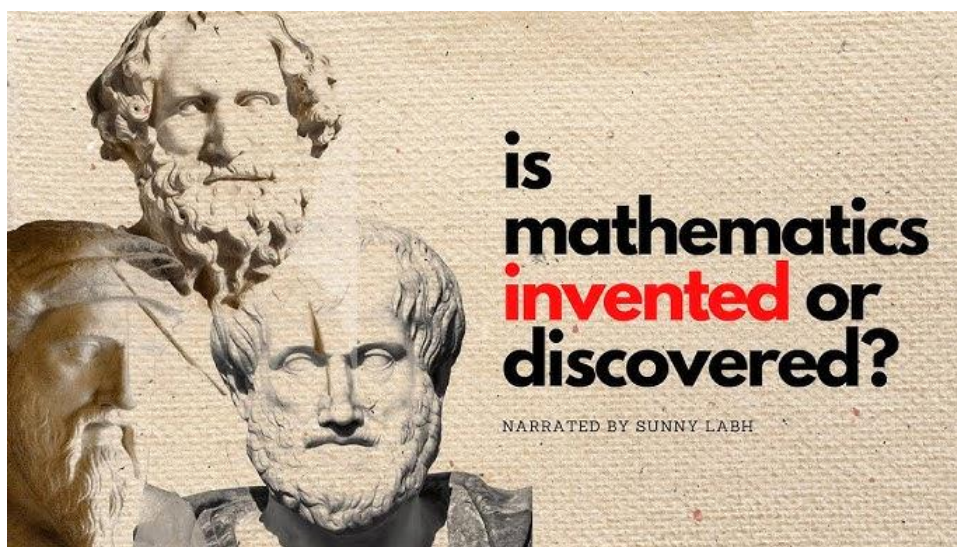




- **Application**

Calculus is applied in many areas of life. It can be used to model systems where there is change.

Examples of the applications of calculus in scientific fields are space exploration, telecommunications systems, computer science, engineering, medicine, pharmacology, business, meteorology, and music.



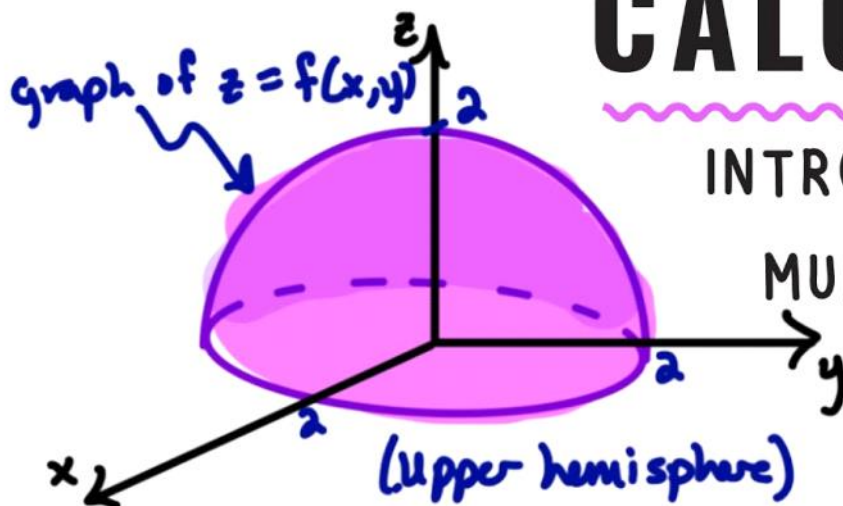
$$f(x, y) = \sqrt{4 - x^2 - y^2}$$

MULTIVARIABLE CALCULUS

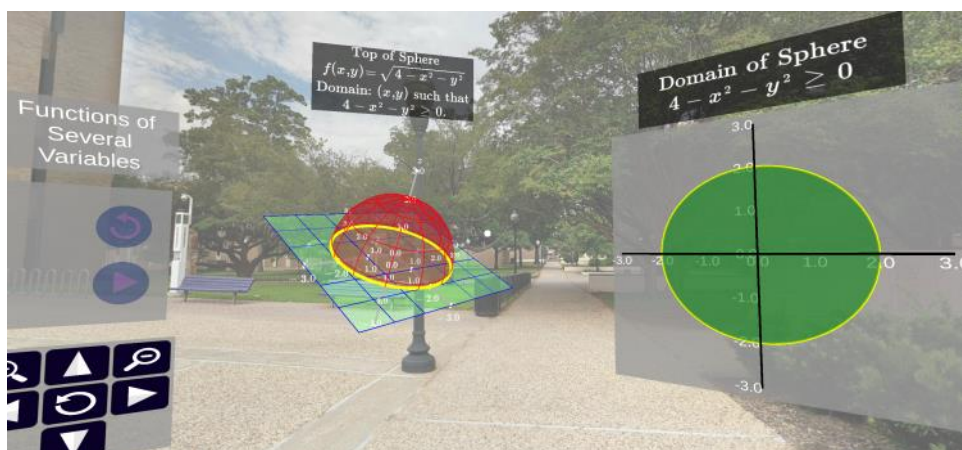
INTRODUCTION TO

MULTIVARIABLE

FUNCTIONS



- Understand the concept of multivariable functions, especially of the form $z = f(x, y)$
- Understand the domain and range of a multivariable functions
- Determine whether a representation/graph can be expressed with one coordinate as a function of the others



Functions arise whenever one quantity depends on another. Consider the following four situations.

- The area A of a circle depends on the radius r of the circle. The rule that connects r and A is given by the equation $A = \pi r^2$. With each positive number r there is associated one value of A , and we say that A is a *function* of r .

- c. The cost C of mailing an envelope depends on its weight w . Although there is no simple formula that connects w and C , the post office has a rule for determining C when w is known.



FIGURE 2
Machine diagram for a function f

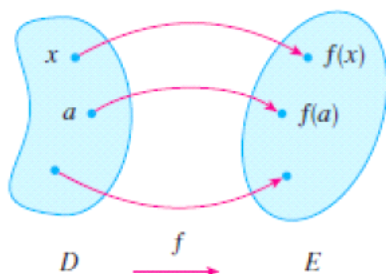


FIGURE 3
Arrow diagram for f

In this section we study functions of two or more variables from four points of view:

- verbally (by a description in words)
- numerically (by a table of values)
- algebraically (by an explicit formula)
- visually (by a graph or level curves)

Example: Beef Consumption

Suppose you are a beef producer and you want to know how much beef people will buy. This depends on how much money people have and on the price of beef. The consumption of beef, C (in pounds per week per household) is a function of household income, I (in thousands of dollars per year), and the price of beef, p (in dollars per pound). In function notation, we write:

$$C = f(I, p).$$

Table 12.1 contains values of this function. Values of p are shown across the top, values of I are down the left side, and corresponding values of $f(I, p)$ are given in the table.¹ For example, to find the value of $f(40, 3.50)$, we look in the row corresponding to $I = 40$ under $p = 3.50$, where we find the number 4.05. Thus,

$$f(40, 3.50) = 4.05.$$

This means that, on average, if a household's income is \$40,000 a year and the price of beef is \$3.50/lb, the family will buy 4.05 lbs of beef per week.

Table 12.1 Quantity of beef bought (pounds/household/week)

		Price of beef (\$/lb)			
		3.00	3.50	4.00	4.50
Household income per year, I (\$1000)	20	2.65	2.59	2.51	2.43
	40	4.14	4.05	3.94	3.88
	60	5.11	5.00	4.97	4.84
	80	5.35	5.29	5.19	5.07
	100	5.79	5.77	5.60	5.53

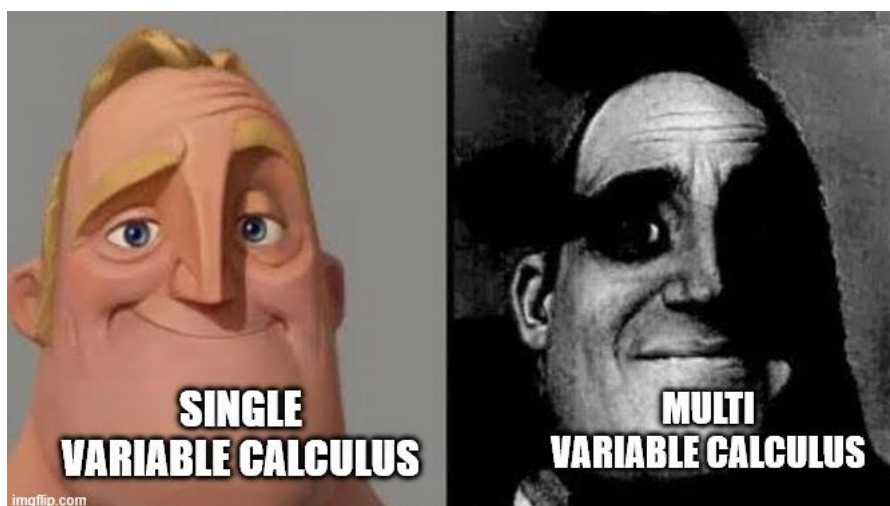
Example 3 A cylinder with closed ends has radius r and height h . If its volume is V and its surface area is A , find formulas for the functions $V = f(r, h)$ and $A = g(r, h)$.

Solution Since the area of the circular base is πr^2 , we have

$$V = f(r, h) = \text{Area of base} \cdot \text{Height} = \pi r^2 h.$$

The surface area of the side is the circumference of the bottom, $2\pi r$, times the height h , giving $2\pi r h$. Thus,

$$A = g(r, h) = 2 \cdot \text{Area of base} + \text{Area of side} = 2\pi r^2 + 2\pi r h.$$



Visualizing a Function of Two Variables Using a Graph

For a function of one variable, $y = f(x)$, the graph of f is the set of all points (x, y) in 2-space such that $y = f(x)$. In general, these points lie on a curve in the plane. When a computer or calculator graphs f , it approximates by plotting points in the xy -plane and joining consecutive points by line segments. The more points, the better the approximation.

Now consider a function of two variables.

The **graph** of a function of two variables, f , is the set of all points (x, y, z) such that $z = f(x, y)$. In general, the graph of a function of two variables is a surface in 3-space.

Plotting the Graph of the Function $f(x, y) = x^2 + y^2$

<https://www.desmos.com/3d>

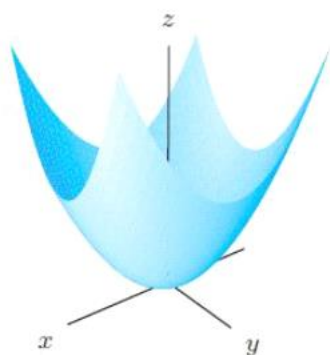


Figure 12.13: Graph of $f(x, y) = x^2 + y^2$ for $-3 \leq x \leq 3, -3 \leq y \leq 3$

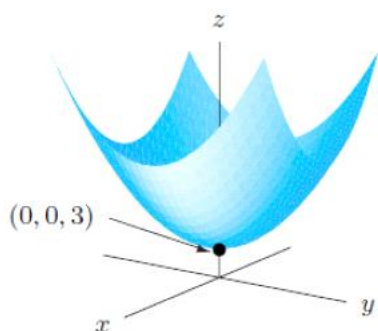


Figure 12.14: Graph of $g(x, y) = x^2 + y^2 + 3$

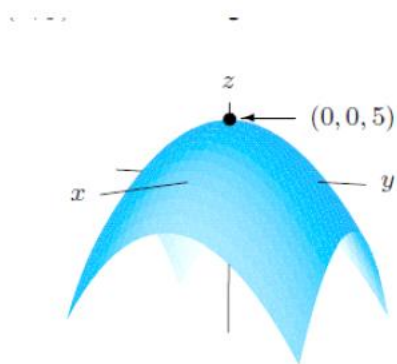


Figure 12.15: Graph of $h(x, y) = 5 - x^2 - y^2$

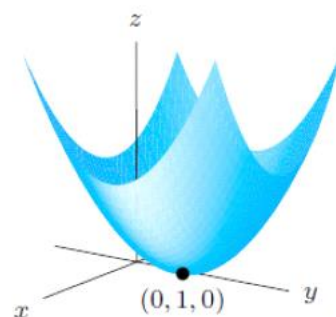


Figure 12.16: Graph of $k(x, y) = x^2 + (y - 1)^2$

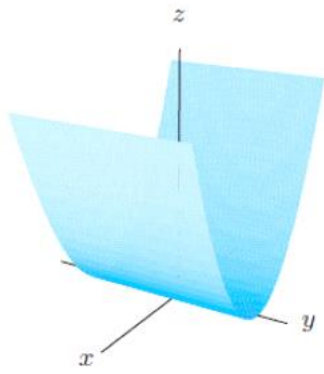


Figure 12.25: A parabolic cylinder $z = x^2$

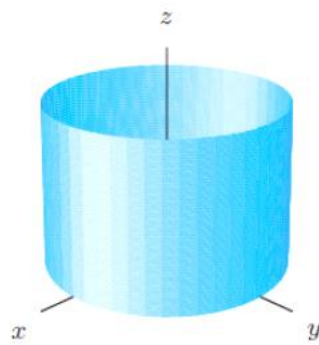
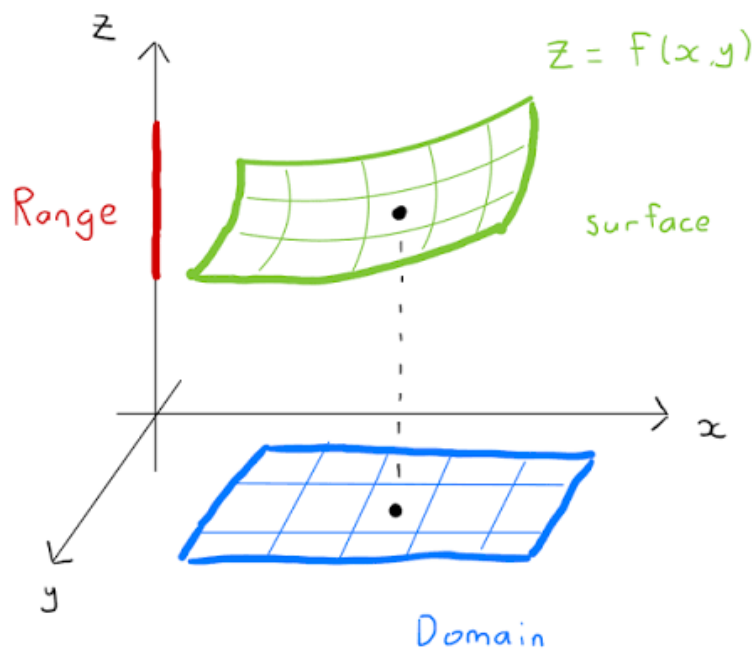


Figure 12.26: Circular cylinder $x^2 + y^2 = 1$



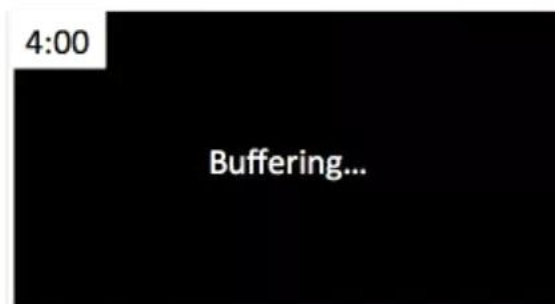
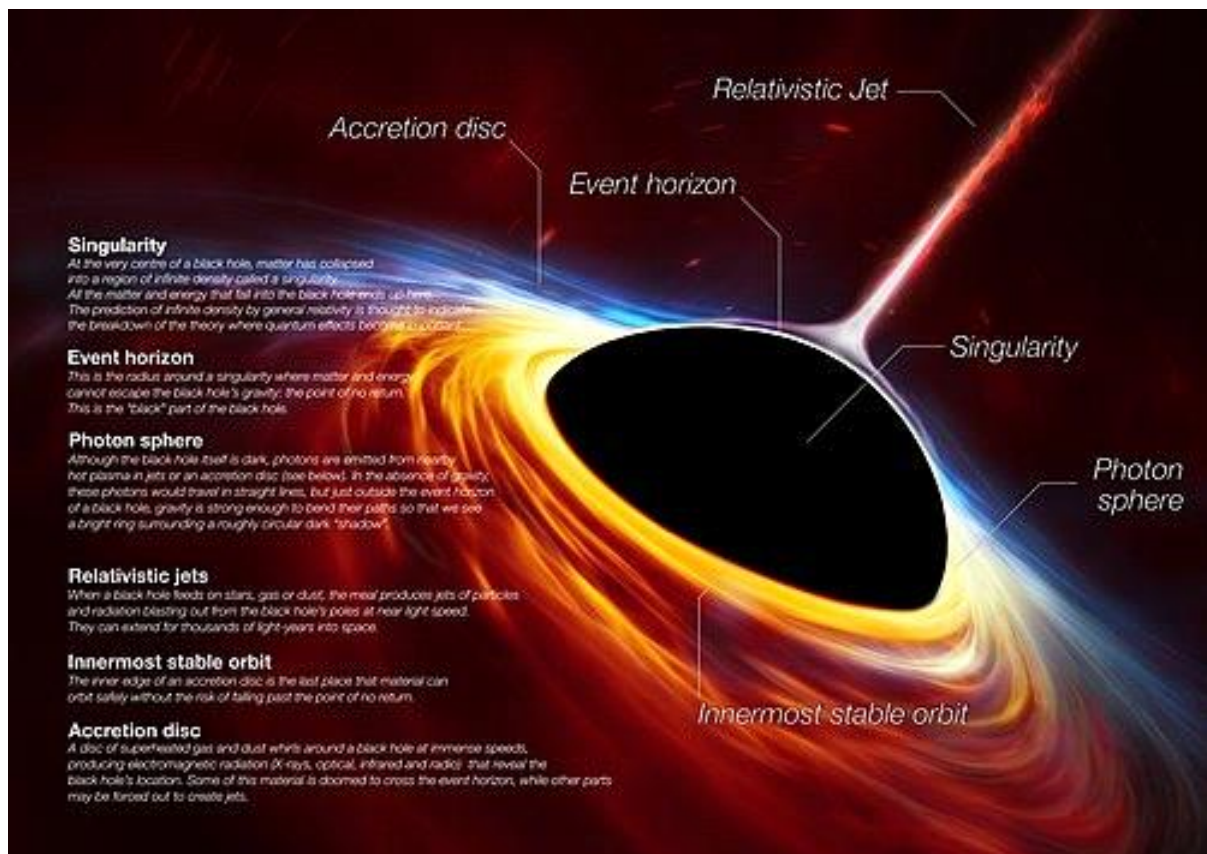
LIMITS

WHAT IS A LIMIT?

- ~ a "limit" can be understood as a wall.
- a value that you can get closer and closer and closer to but never achieve.



- limits are the same; you can get a number that is closer and closer but never that number because it does not exist.



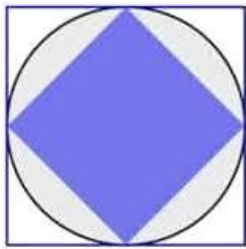


Mathematicians:	3.141592653589793238462643383279502 88419716939937510582097494459230191 64062862089986280348253421170879821 48896513282306647093844609550582231 72535940812848111745028416270193852 11053596446229489549303819644288109 75665933446128475648233786783163271 30190914564854692346034861045432644 82133936072602491412737245870066063 35508174881520926942829254081715364 36789259036001133053854882046652138 41469519415114094330572703657595919 53092186117381932611793105118548074
Normal People:	$\pi = 3.14$
Engineers:	

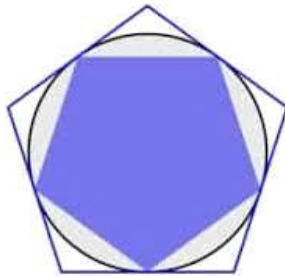
The Fundamental Theorem of Engineering

- $\pi = 3$
- $e = 2$
- $\pi = e$
- $\sin(x) = x$
- $\cos(x) = 1$
- If it's close enough,
then it's good enough

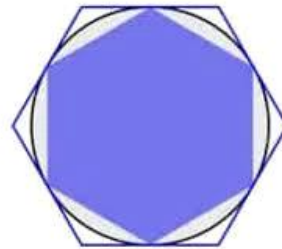




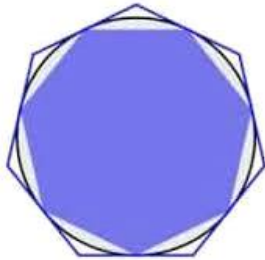
n=4



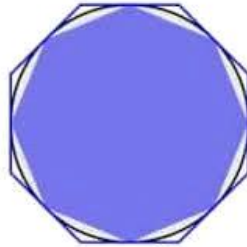
n=5



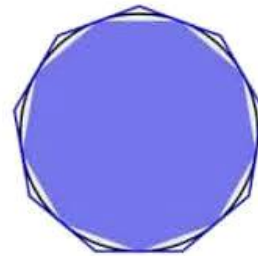
n=6



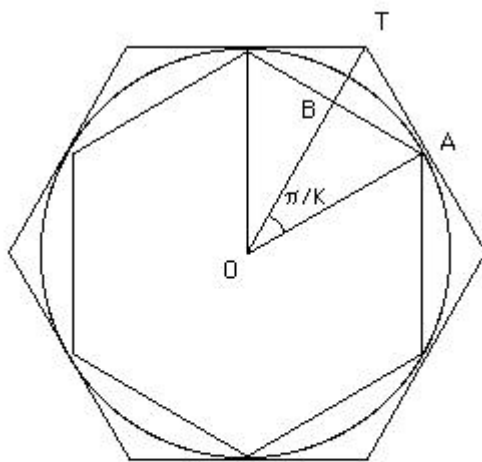
n=7



n=8



n=9



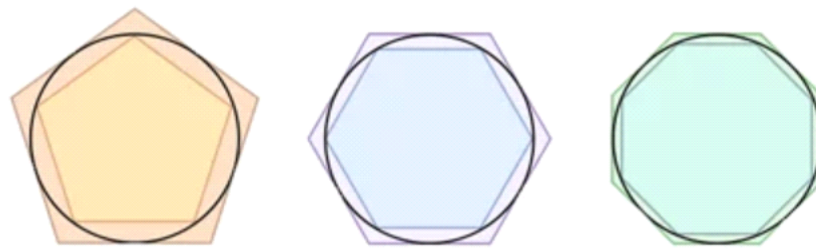
$$\begin{aligned} OA &= 1 \\ AB &= \sin(\pi/K) \\ AT &= \tan(\pi/K) \\ \text{where } K &= 3 \times 2^{n-1} \end{aligned}$$

Archimedes figured out that pi had a range of

$$3\frac{10}{71} < \pi < 3\frac{1}{7}$$

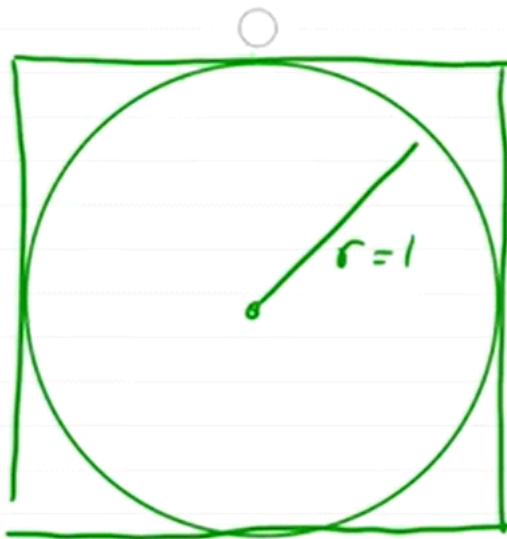
using a process like this:

More Sides = Better “Circle”



Archimedes (350 BC) Approximate π .

Area: $\pi r^2 = \pi$

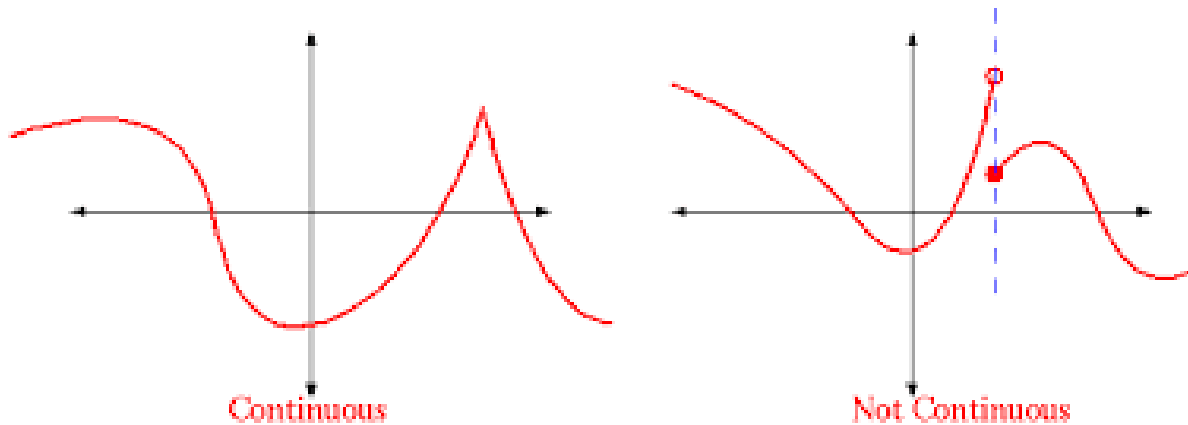


Upper bound on π is 4.

Lower bound on π is 2.

$$2 < \pi < 4 \quad \pi \approx 3.$$

CONTINUITY



$f : X \rightarrow Y$ is continuous if
it can be drawn without
lifting pen from paper

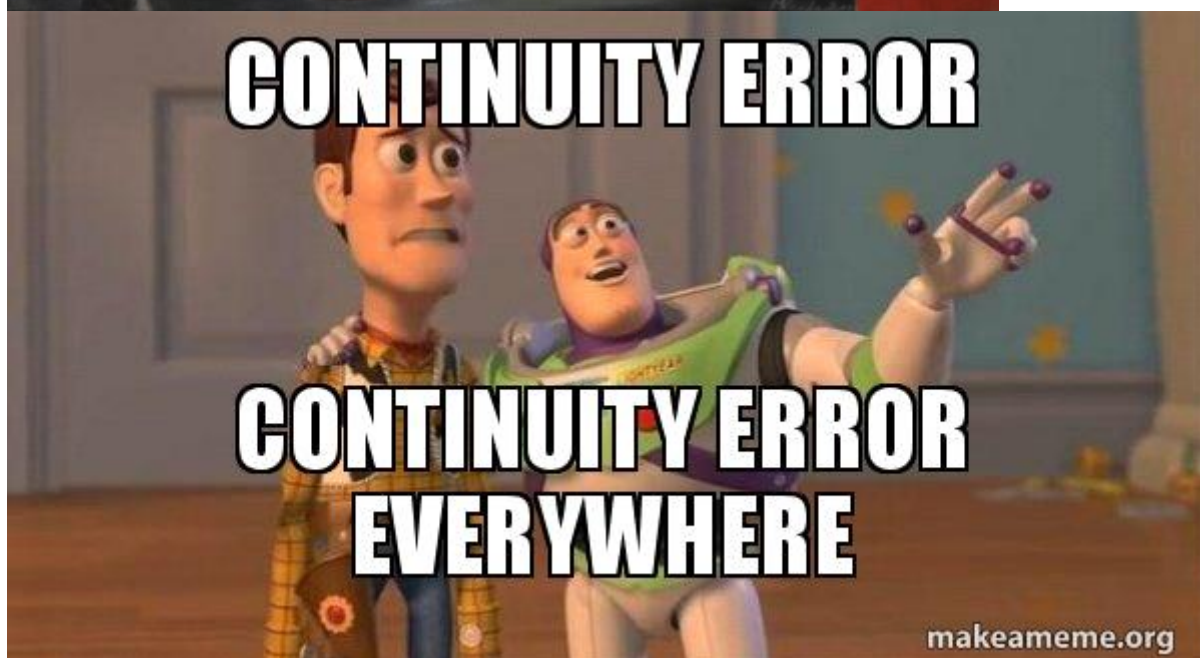


$f : X \rightarrow Y$ is continuous if
 $\forall \varepsilon > 0, \exists \delta > 0$
 $|x - c| < \delta \implies |f(x) - f(c)| < \varepsilon$



$f : X \rightarrow Y$ is continuous if
 $\forall V \in \mathcal{T}_Y, f^{-1}(V) \in \mathcal{T}_X$





Core Reasons for Engineering

- **Predictable Modeling:** Real physical quantities like temperature, pressure, and velocity change without teleporting from one value to another. Continuous functions match this natural reality.
- **Calculus and Rates of Change:** Derivatives and integrals—used to find speed, acceleration, and total work—only work smoothly if a function is continuous.  StudyPug
- **Conservation Laws:** Engineering relies on the [Continuity Equation](#) to prove that mass, energy, and momentum cannot be created or destroyed out of nowhere.  Wikipedia
- **System Stability:** Control systems in robotics or aerospace need continuous signals to prevent sudden shocks that could break physical machinery.

If you'd like, I can focus on a **specific branch of engineering** (like mechanical, civil, or electrical) to show you exact formulas and examples where continuity matters.